

Analysing vehicles registration in 1Q 2026. The dataset was obtain from Department of Statistic Malaysia. This analysis is to study the market pattern, preferences based on the car makers, types, fuel etc. using several types of exploratory data analysis techniques and lastly, to compare how well the registrations is compare to 2025 on the same period.

```
In [ ]: ## Loading up the required R Library packages
```

```
library(tidyverse)
library(janitor)
library(skimr)
library(scales)
library(uuid)
library(lubridate)
```

```
In [2]: ## Loading dataset
```

```
df_jpj <- read.csv("cars_2026.csv")
```

```
In [3]: ## Checking the dataset
```

```
glimpse(df_jpj)
```

Rows: 270,314

Columns: 7

```
$ date_reg <chr> "2026-01-01", "2026-01-01", "2026-01-01", "2026-01-01", "20
26...
$ type      <chr> "motokar", "motokar", "jip", "jip", "pick_up", "jip", "ji
p", ...
$ maker     <chr> "BMW", "BMW", "Chery", "Chery", "Ford", "GAC", "Great Wal
l", ...
$ model     <chr> "7 Series", "7 Series", "iCaur V23", "Tiggo", "Ranger", "GS
3"...
$ colour    <chr> "black", "grey", "orange", "grey", "black", "silver", "whit
e"...
$ fuel      <chr> "hybrid_petrol", "hybrid_petrol", "electric", "petrol", "pe
tr...
$ state     <chr> "Johor", "Rakan Niaga", "Rakan Niaga", "Rakan Niaga", "Joho
r"...
```

```
In [4]: head(df_jpj)
        tail(df_jpj)
```

A data.frame: 6 × 7

	date_reg	type	maker	model	colour	fuel	state
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	2026-01-01	motokar	BMW	7 Series	black	hybrid_petrol	Johor
2	2026-01-01	motokar	BMW	7 Series	grey	hybrid_petrol	Rakan Niaga
3	2026-01-01	jip	Chery	iCaur V23	orange	electric	Rakan Niaga
4	2026-01-01	jip	Chery	Tiggo	grey	petrol	Rakan Niaga
5	2026-01-01	pick_up	Ford	Ranger	black	petrol	Johor
6	2026-01-01	jip	GAC	GS3	silver	petrol	Rakan Niaga

A data.frame: 6 × 7

	date_reg	type	maker	model	colour	fuel	state
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
270309	2026-04-30	jip	Audi	Q7	black	hybrid_petrol	Pulau Pinang
270310	2026-04-30	jip	Audi	Q7	blue	hybrid_petrol	Rakan Niaga
270311	2026-04-30	jip	Audi	Q7	white	hybrid_petrol	Rakan Niaga
270312	2026-04-30	jip	Audi	Q8	white	petrol	Rakan Niaga
270313	2026-04-30	motokar	Audi	RS3	grey	petrol	W.P. Kuala Lumpur
270314	2026-04-30	motokar	Audi	RS6	red	petrol	Selangor

Data was from 1 January 2026 till 30 April 2026. Its consist of 7 columns and 270,314 rows. From the summary we can see one column has a wrong data type.

```
In [5]: ## converting date_reg to date data type
df_jpj$date_reg <- ymd(df_jpj$date_reg)
```

```
In [6]: class(df_jpj$date_reg)
str(df_jpj)
```

'Date'

```
'data.frame': 270314 obs. of 7 variables:
 $ date_reg: Date, format: "2026-01-01" "2026-01-01" ...
 $ type    : chr  "motokar" "motokar" "jip" "jip" ...
 $ maker   : chr  "BMW" "BMW" "Chery" "Chery" ...
 $ model   : chr  "7 Series" "7 Series" "iCaur V23" "Tiggo" ...
 $ colour  : chr  "black" "grey" "orange" "grey" ...
 $ fuel    : chr  "hybrid_petrol" "hybrid_petrol" "electric" "petrol" ...
 $ state   : chr  "Johor" "Rakan Niaga" "Rakan Niaga" "Rakan Niaga" ...
```

Now we have the correct data type for date_reg column. It's time to investigate the data set

```
In [7]: ## Checking the data integrity

## Summarise the data using skimr

skim(df_jpj)
```

	skim_type	skim_variable	n_missing	complete_rate	Date.min	Date.max	Date.i
	<chr>	<chr>	<int>	<dbl>	<date>	<date>	
1	Date	date_reg	0	1	2026-01-01	2026-04-30	2026
2	character	type	0	1	NA	NA	
3	character	maker	0	1	NA	NA	
4	character	model	0	1	NA	NA	
5	character	colour	0	1	NA	NA	
6	character	fuel	0	1	NA	NA	
7	character	state	0	1	NA	NA	

```
In [8]: ## checking for null values
colSums(is.na(df_jpj))
```

date_reg: 0 type: 0 maker: 0 model: 0 colour: 0 fuel: 0 state: 0

No null value found.

```
In [9]: ## checking for duplication
sum(duplicated(df_jpj))
```

219161

Wait what?

```
In [ ]: df_jpj[duplicated(df_jpj), ]
```

Okay that's lot of duplicated data. However, logically speaking, it can't be a duplication of registrations since in this data set, I realise there is no unique key to identify the transaction. Two similar maker, model, type, or coloured registered cars could have been two separate entity registering the same exact car model, type, colours, in the same state. So it can't be a duplication.

So I've decided to create the new column as transaction ID to differentiate the registration.

```
In [11]: ## Create transaction ID column

df_jpj <- df_jpj %>%
  mutate(
```

```

regID = paste0(
  "reg_", format(date_reg, "%Y%m%d"), "_",
  sprintf("%04d", row_number())
)

```

In [12]:

```

head(df_jpj)
tail(df_jpj)

```

A data.frame: 6 × 8

	date_reg	type	maker	model	colour	fuel	state	regID
	<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	2026-01-01	motokar	BMW	7 Series	black	hybrid_petrol	Johor	reg_20260101_0000
2	2026-01-01	motokar	BMW	7 Series	grey	hybrid_petrol	Rakan Niaga	reg_20260101_0001
3	2026-01-01	jip	Chery	iCaur V23	orange	electric	Rakan Niaga	reg_20260101_0002
4	2026-01-01	jip	Chery	Tiggo	grey	petrol	Rakan Niaga	reg_20260101_0003
5	2026-01-01	pick_up	Ford	Ranger	black	petrol	Johor	reg_20260101_0004
6	2026-01-01	jip	GAC	GS3	silver	petrol	Rakan Niaga	reg_20260101_0005

A data.frame: 6 × 8

	date_reg	type	maker	model	colour	fuel	state	
	<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	
270309	2026-04-30	jip	Audi	Q7	black	hybrid_petrol	Pulau Pinang	reg_20260430_0000
270310	2026-04-30	jip	Audi	Q7	blue	hybrid_petrol	Rakan Niaga	reg_20260430_0001
270311	2026-04-30	jip	Audi	Q7	white	hybrid_petrol	Rakan Niaga	reg_20260430_0002
270312	2026-04-30	jip	Audi	Q8	white	petrol	Rakan Niaga	reg_20260430_0003
270313	2026-04-30	motokar	Audi	RS3	grey	petrol	W.P. Kuala Lumpur	reg_20260430_0004
270314	2026-04-30	motokar	Audi	RS6	red	petrol	Selangor	reg_20260430_0005

Refining the regID column to make the id is reset by date.

In [13]:

```

df_jpj <- df_jpj %>%
  group_by(date_reg) %>%
  mutate(

```

```

regID = paste0(
  "reg_", format(date_reg, "%Y%m%d"), "_",
  sprintf("%04d", row_number())
) %>%
ungroup()

```

```

In [14]: head(df_jpj)
         tail(df_jpj)

```

A tibble: 6 × 8

date_reg	type	maker	model	colour	fuel	state	regID
<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
2026-01-01	motokar	BMW	7 Series	black	hybrid_petrol	Johor	reg_20260101_0001
2026-01-01	motokar	BMW	7 Series	grey	hybrid_petrol	Rakan Niaga	reg_20260101_0002
2026-01-01	jip	Chery	iCaur V23	orange	electric	Rakan Niaga	reg_20260101_0003
2026-01-01	jip	Chery	Tiggo	grey	petrol	Rakan Niaga	reg_20260101_0004
2026-01-01	pick_up	Ford	Ranger	black	petrol	Johor	reg_20260101_0005
2026-01-01	jip	GAC	GS3	silver	petrol	Rakan Niaga	reg_20260101_0006

A tibble: 6 × 8

date_reg	type	maker	model	colour	fuel	state	regID
<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
2026-04-30	jip	Audi	Q7	black	hybrid_petrol	Pulau Pinang	reg_20260430_1464
2026-04-30	jip	Audi	Q7	blue	hybrid_petrol	Rakan Niaga	reg_20260430_1464
2026-04-30	jip	Audi	Q7	white	hybrid_petrol	Rakan Niaga	reg_20260430_1464
2026-04-30	jip	Audi	Q8	white	petrol	Rakan Niaga	reg_20260430_1464
2026-04-30	motokar	Audi	RS3	grey	petrol	W.P. Kuala Lumpur	reg_20260430_1465
2026-04-30	motokar	Audi	RS6	red	petrol	Selangor	reg_20260430_1465

Let' check whether the reset number by date is working or not

```

In [15]: tail(which(df_jpj$date_reg == as.Date("2026-01-01")), 1)
         head(which(df_jpj$date_reg == as.Date("2026-01-02")), 1)

```

2584

2585

```
In [16]: df_jpj %>% slice(2584:2585)
```

A tibble: 2 × 8

date_reg	type	maker	model	colour	fuel	state	regID
<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
2026-01-01	motokar	Toyota	Yaris	red	petrol	Rakan Niaga	reg_20260101_2584
2026-01-02	jip	Audi	RS Q3	grey	petrol	Rakan Niaga	reg_20260102_0001

OK the ID column is working as intended

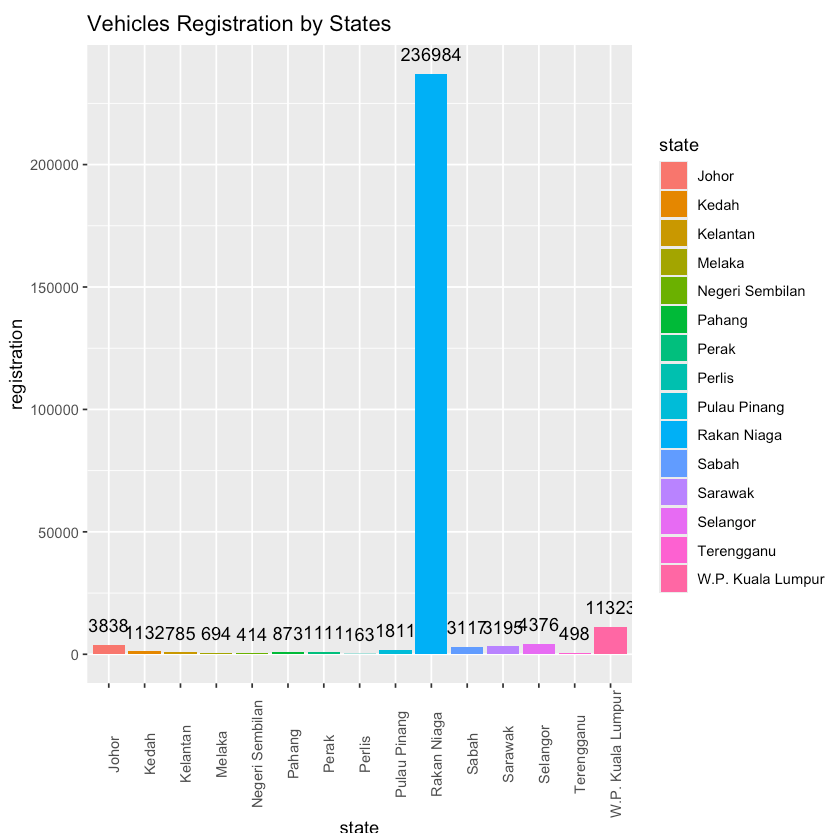
Let's save the new cleaned dataset for future explorations.

```
In [17]: df_jpj %>% write_csv("jpj_1Q2026.csv")
```

Now lets dive into the data and what it's tell

```
In [18]: ## Number of vehicles registered by states
```

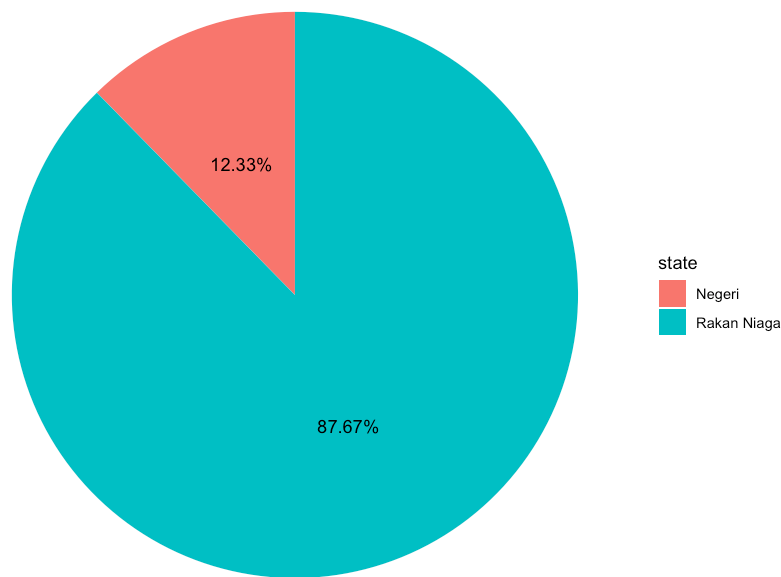
```
df_jpj %>%  
  count (  
    state, name = "registration"  
  ) %>%  
  ggplot(aes(state, registration, fill = state)) +  
  geom_col() +  
  geom_text(aes(label = registration), vjust = -1 ) +  
  theme(axis.text.x = element_text(angle = 90)) +  
  ggtitle("Vehicles Registration by States")
```



There were 236984 vehicles were registered through Rakan Niaga. From the documentation from the JPJ, Rakan Niaga means the vehicles registrations from the authorised third party services.

```
In [19]: df_jpj %>%
  mutate(
    state = if_else(
      state == "Rakan Niaga",
      "Rakan Niaga",
      "Negeri"
    ) %>%
    count(state, name = 'registrations') %>%
    mutate(
      pct = round(registrations / sum(registrations) * 100, 2)
    ) %>%
    ggplot(aes(x = "", y = pct, fill = state)) +
    geom_col() +
    coord_polar(theta = "y") +
    geom_text(aes(label = paste0(pct, "%")), position = position_just(x = 1)) +
    ggtitle("Car Registration by States vs Rakan Niaga") +
    theme_void()
```

Car Registration by States vs Rakan Niaga



Registration from Rakan Niaga alone was about 87% from overall registration. It is impossible to obtain accurate information of breakdown the registration from the states since it could skewed the numbers. Therefore I suggest to drop to study the registration from the states extensively as I don't want any skewed result.

But we can study the data from other areas.

```
In [20]: ## Create new data frame
```

```
df_jpj26 <- df_jpj
```

```
In [21]: ## number of cars registered
```

```
count(df_jpj26, name = "total registration")
```

A tibble: 1 × 1

total registration

<int>

270314

There were 270314 vehicles were registered for 1 January 2026 till 30 April 2026.

```
In [22]: ## Comparing the numbers between national and non national brands
```

```
market_share <- df_jpj26 %>%  
  summarise(
```



```
national = sum(maker %in% c("Perodua", "Proton")),  
'non national' = sum(!maker %in% c("Perodua", "Proton")),  
) %>%  
  pivot_longer(  
    everything(), names_to = "group", values_to = "total_reg  
  ) %>%  
  mutate(  
    pct = round(total_register / sum(total_register) * 100,  
  )
```

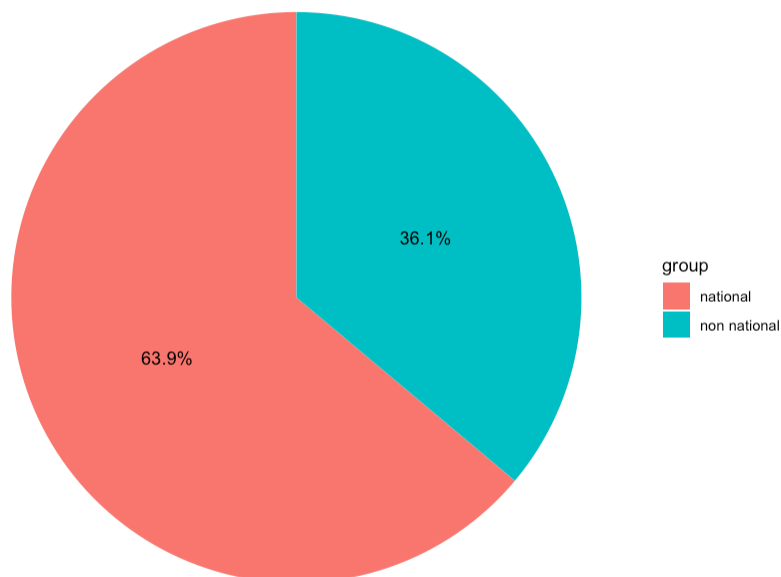
market_share

A tibble: 2 × 3

group	total_register	pct
<chr>	<int>	<dbl>
national	172718	63.9
non national	97596	36.1

```
In [23]: ggplot(market_share, aes(x = "", y = pct, fill = group)) +  
  geom_col() +  
  coord_polar(theta = "y") +  
  geom_text(aes(label = paste0(pct, "%")), position = position_stack(v  
  theme_void() +  
  ggtitle("Market Share National vs Non National Brands")
```

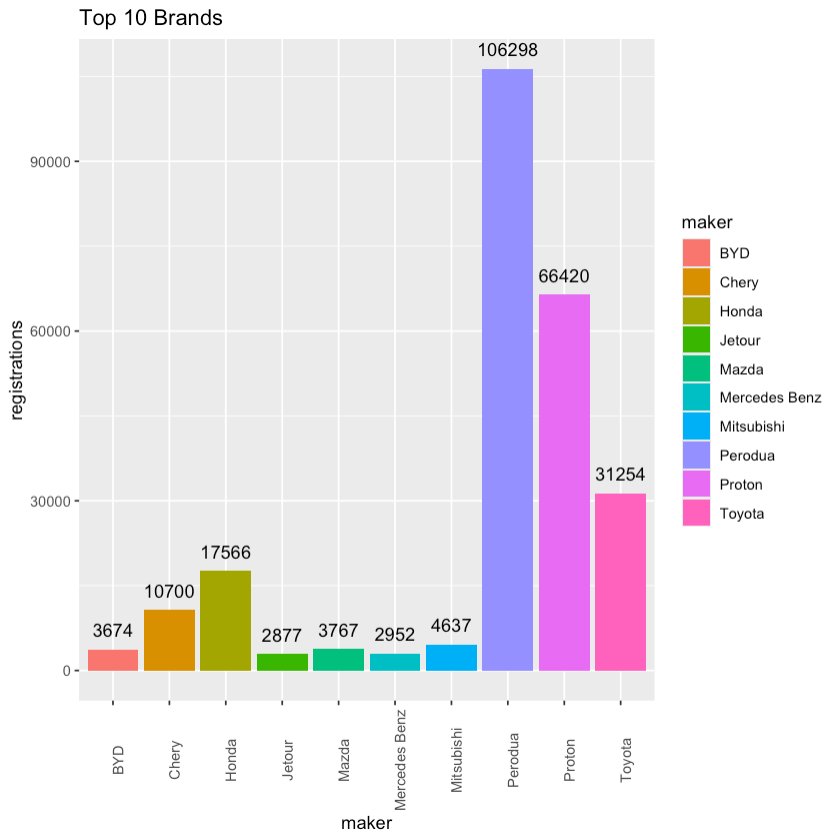
Market Share National vs Non National Brands



Proton and Perodua, being a national car makers dominated the market.

In [24]: *## Top 10 vehicles maker*

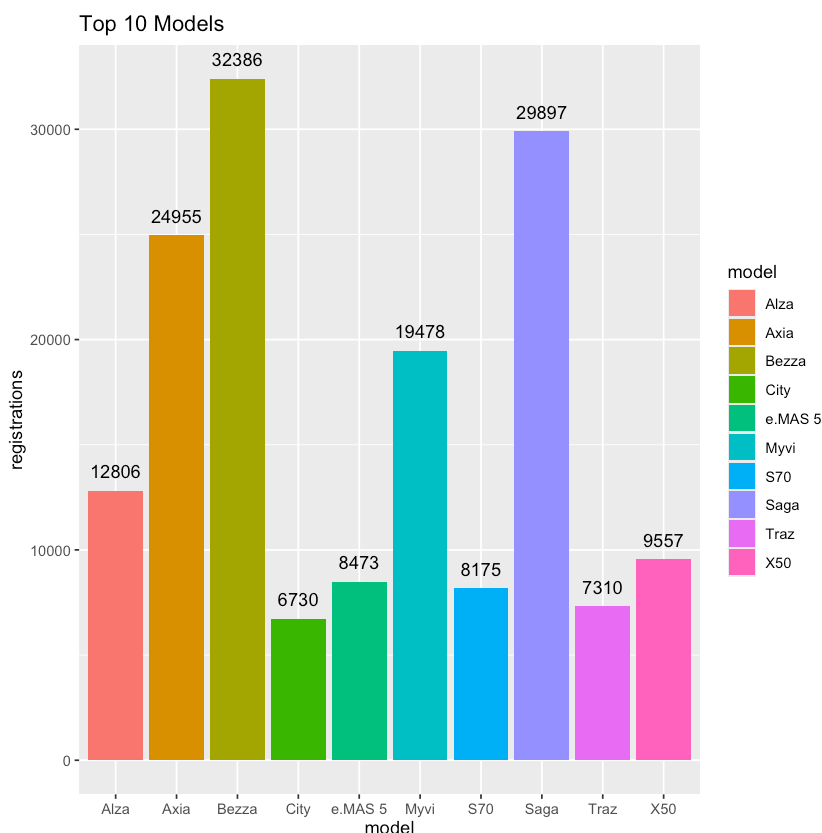
```
df_jpj26 %>%
  count(maker, name = "registrations") %>%
  slice_max(registrations, n = 10) %>%
  ggplot(aes(maker, registrations, fill = maker)) +
  geom_col() +
  geom_text(aes(label = registrations), vjust = -1) +
  theme(axis.text.x = element_text(angle = 90)) +
  ggtitle("Top 10 Brands")
```



Perodua vehicles top the list. Being a mass production car, Perodua also are well known for its cheap, longevity, reliability, ease to maintain, spare part availability and good resale value.

In [25]: *## Top 10 model registered*

```
df_jpj26 %>%
  count(model, name = "registrations") %>%
  slice_max(registrations, n = 10) %>%
  ggplot(aes(model, registrations, fill = model)) +
  geom_col() +
  geom_text(aes(label = registrations), vjust = -1) +
  ggtitle("Top 10 Models")
```



Perodua Bezza and Proton Saga takes the top two spots. Being an affordable sedan with large boot, reliable, and ease to maintain, it is attractive for lower income group to have these model as their everyday, family car. Both models also attractive to being used as e-hailing car since it have large boot suitable to carry passengers and tourists around.

In [26]: *## Number 1 registered vehicle registered by state excluding the Rakan Niaga*

```
df_jpj26 %>%
  filter (state != 'Rakan Niaga' ) %>%
  count(state, model, name = 'registrations') %>%
  group_by(state) %>%
  slice_max(registrations, n = 1) %>%
  ungroup()
```

A tibble: 14 × 3

state	model	registrations
<chr>	<chr>	<int>
Johor	Alphard	570
Kedah	Alphard	94
Kelantan	Hilux	89
Melaka	Alphard	74
Negeri Sembilan	Hilux	63
Pahang	Hilux	198
Perak	Hilux	217
Perlis	Hilux	19
Pulau Pinang	Alphard	259
Sabah	Hilux	1989
Sarawak	Hilux	1872
Selangor	Hilux	697
Terengganu	Hilux	69
W.P. Kuala Lumpur	Alphard	1496

Surprise to learnt Toyota Hilux is a choice of many. With a reputation of being rugged and reliable pickup truck, it does have its own appeal especially for business owner.

```
In [27]: ## Number of vehicles registered by states excluding the Rakan Niaga

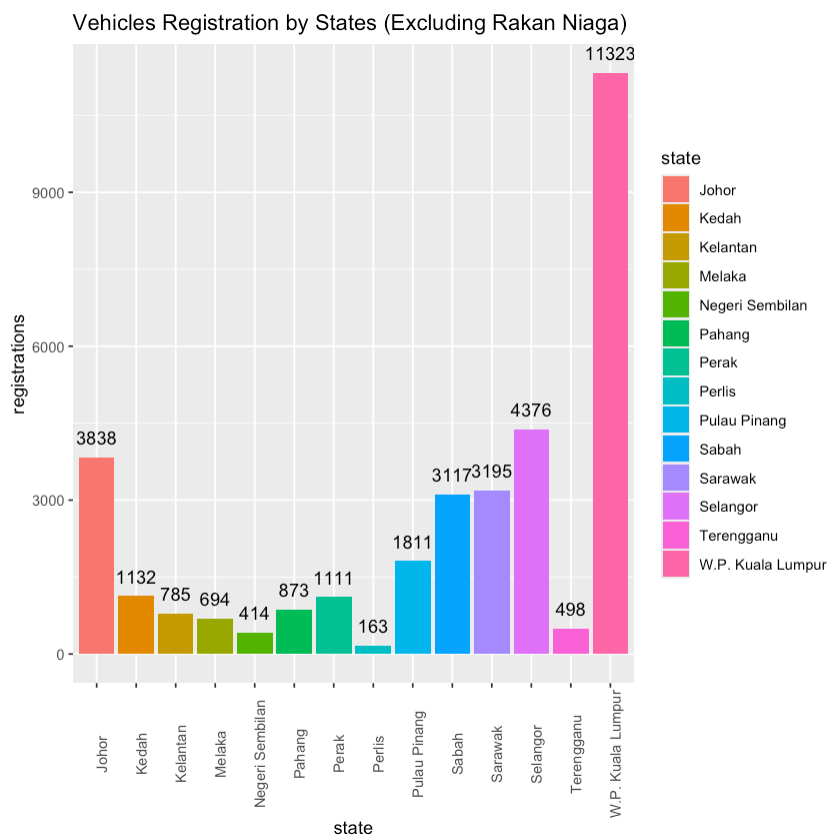
df_statepct26 <- df_jpj26 %>%
  filter(state != 'Rakan Niaga') %>%
  count(state, name = "total_registrations") %>%
  mutate(
    pct = round(total_registrations / sum(total_registrations) *
  )

df_statepct26

df_jpj26 %>%
  filter(state != 'Rakan Niaga') %>%
  count (
    state, name = "registrations"
  ) %>%
  ggplot(aes(state, registrations, fill = state)) +
  geom_col() +
  geom_text(aes(label = registrations), vjust = -1 ) +
  theme(axis.text.x = element_text(angle = 90)) +
  ggtitle("Vehicles Registration by States (Excluding Rakan Niaga")
```

A tibble: 14 × 3

state	total_registrations	pct
<chr>	<int>	<dbl>
Johor	3838	11.52
Kedah	1132	3.40
Kelantan	785	2.36
Melaka	694	2.08
Negeri Sembilan	414	1.24
Pahang	873	2.62
Perak	1111	3.33
Perlis	163	0.49
Pulau Pinang	1811	5.43
Sabah	3117	9.35
Sarawak	3195	9.59
Selangor	4376	13.13
Terengganu	498	1.49
W.P. Kuala Lumpur	11323	33.97

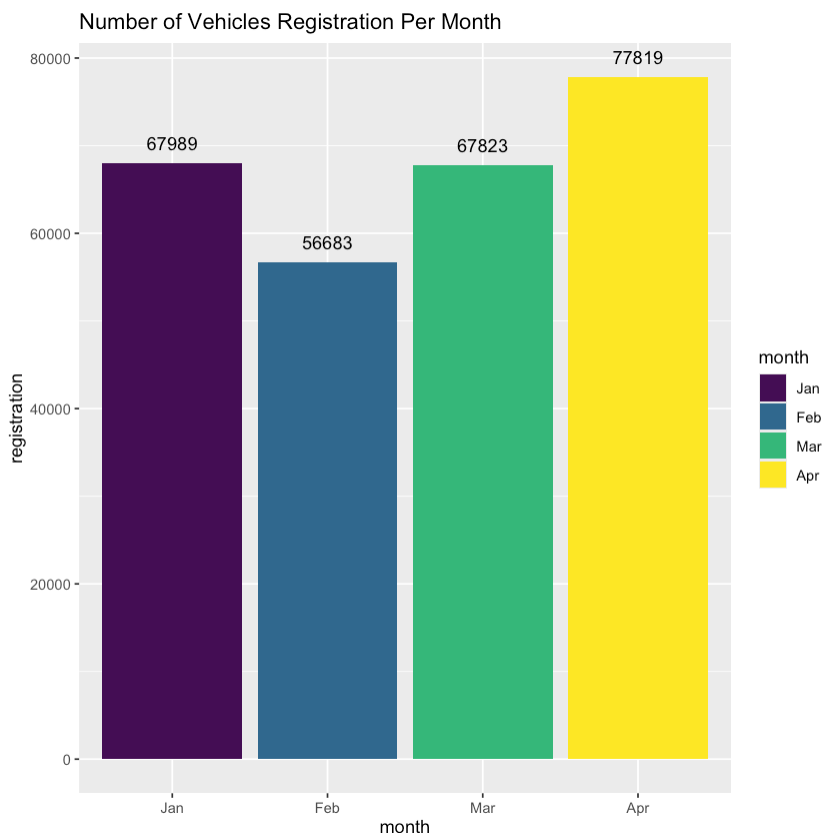


WP Kuala Lumpur topped the list. No suprise, being the country capital state and active

economics centre people have more buying power here, therefore can afford to have a cars.

In [28]: `## Cars registered per month`

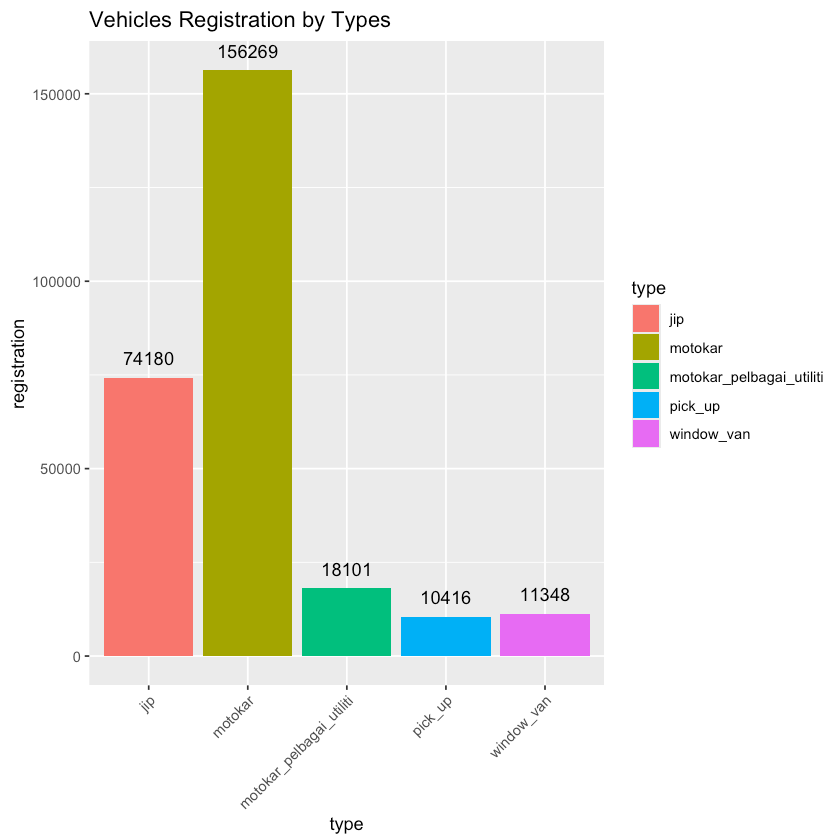
```
df_jpj26 %>%
  mutate(
    month = month(date_reg, label = TRUE) %>%
    group_by(month) %>%
    summarise('registration' = sum(n())) %>%
    ggplot(aes(month, registration, fill = month)) +
    geom_col() +
    geom_text(aes(label = registration), vjust = -1) +
    ggtitle("Number of Vehicles Registration Per Month")
```



There is a dipped in registration in February which is coincide with Ramadan and Hari Raya Aidilfitri, which in generally people are saving up to spend on other festive necessities and buying new car is not really an option during this period.

In [29]: `## By type`

```
df_jpj26 %>%
  count(type, name = "registration", sort = TRUE) %>%
  ggplot(aes(type, registration, fill = type)) +
  geom_col() +
  geom_text(aes(label = registration), vjust = -1) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ggtitle("Vehicles Registration by Types")
```



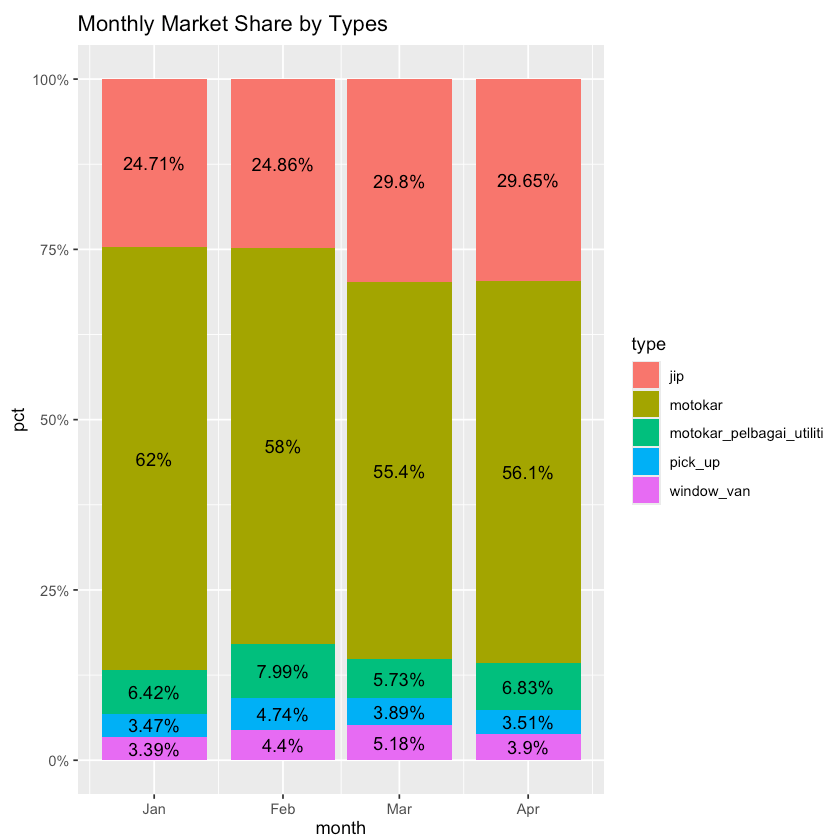
```
In [101] type_month <- df_jpj26 %>%
  mutate(month = floor_date(date_reg, "month")) %>%
  count(month, type) %>%
  group_by(month) %>%
  mutate(pct = round(n / sum(n) * 100, 2)) %>%
  ungroup() %>%
  select(-n)
```

```
In [102] type_month

ggplot(type_month, aes(x = month, y = pct, fill = type)) +
  geom_col() +
  scale_y_continuous(labels = function(x) paste0(x, "%")) +
  geom_text(
    aes(label = paste0(pct, "%")),
    position = position_stack(vjust = 0.5)
  ) +
  ggtitle("Monthly Market Share by Types")
```

A tibble: 20 × 3

month	type	pct
<date>	<chr>	<dbl>
2026-01-01	jip	24.71
2026-01-01	motokar	62.00
2026-01-01	motokar_pelbagai_utiliti	6.42
2026-01-01	pick_up	3.47
2026-01-01	window_van	3.39
2026-02-01	jip	24.86
2026-02-01	motokar	58.00
2026-02-01	motokar_pelbagai_utiliti	7.99
2026-02-01	pick_up	4.74
2026-02-01	window_van	4.40
2026-03-01	jip	29.80
2026-03-01	motokar	55.40
2026-03-01	motokar_pelbagai_utiliti	5.73
2026-03-01	pick_up	3.89
2026-03-01	window_van	5.18
2026-04-01	jip	29.65
2026-04-01	motokar	56.10
2026-04-01	motokar_pelbagai_utiliti	6.83
2026-04-01	pick_up	3.51
2026-04-01	window_van	3.90

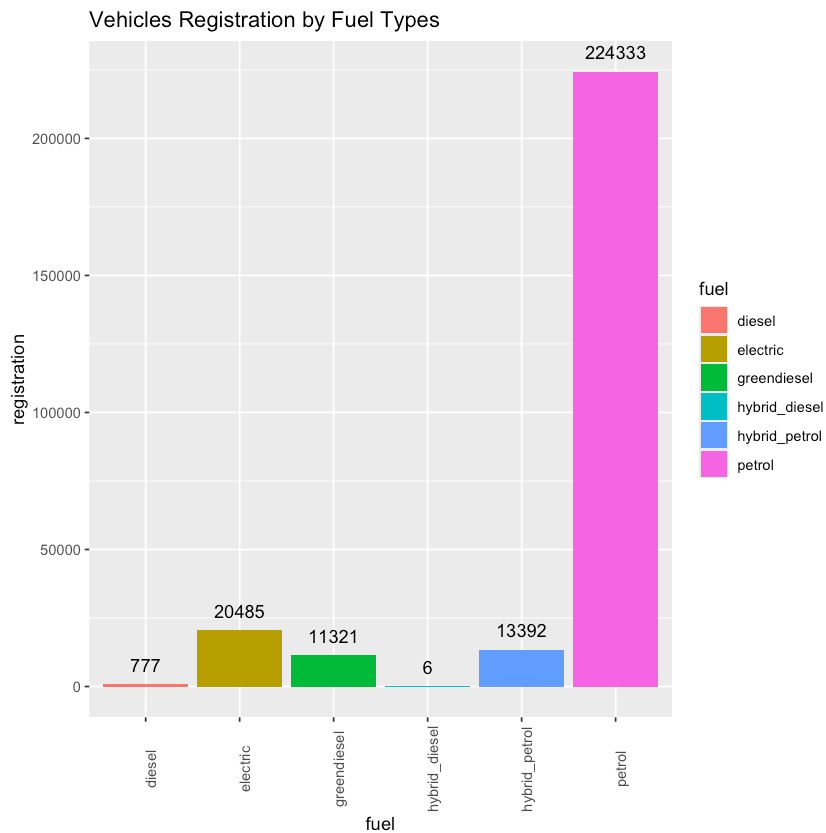


Motokar, or sedan, is still highly preferred by many. Large boot space, low rides, and lightweight body make it more practical than other types for everyday usage.

Motokar pelbagai utiliti, or MPV, recorded high on February, which again coincide with Hari Raya festival where people need to travel with lots of load and MPV suit the best to carry people and belongings.

In [32]: `## By fuel type`

```
df_jpj26 %>%
  count(fuel, name = "registration") %>%
  ggplot(aes(fuel, registration, fill = fuel)) +
  geom_col() +
  geom_text(
    aes(label = registration), vjust = -1) +
  theme(axis.text.x = element_text(angle = 90)) +
  ggtitle("Vehicles Registration by Fuel Types")
```



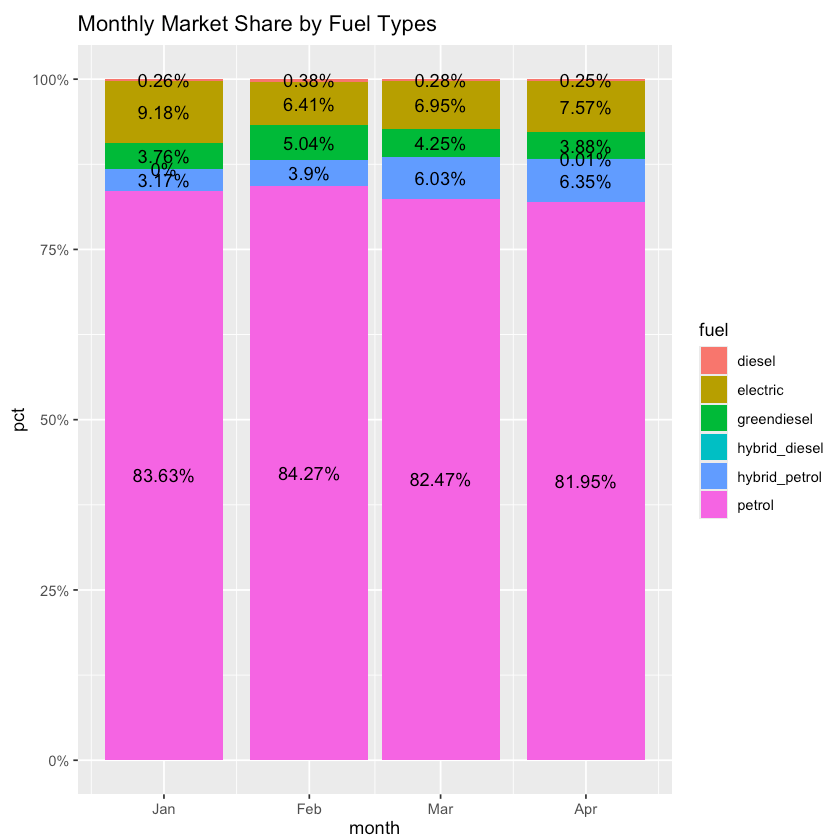
```
In [99]: fuel_month <- df_jpj26 %>%
  mutate(month = floor_date(date_reg, "month")) %>%
  count(month, fuel) %>%
  group_by(month) %>%
  mutate(pct = round(n / sum(n) * 100, 2)) %>%
  ungroup() %>%
  select(-n)
```

```
In [100]: fuel_month

ggplot(fuel_month, aes(x = month, y = pct, fill = fuel)) +
  geom_col() +
  scale_y_continuous(labels = function(x) paste0(x, "%")) +
  geom_text(
    aes(label = paste0(pct, "%")),
    position = position_stack(vjust = 0.5)
  ) +
  ggtitle("Monthly Market Share by Fuel Types")
```

A tibble: 22 × 3

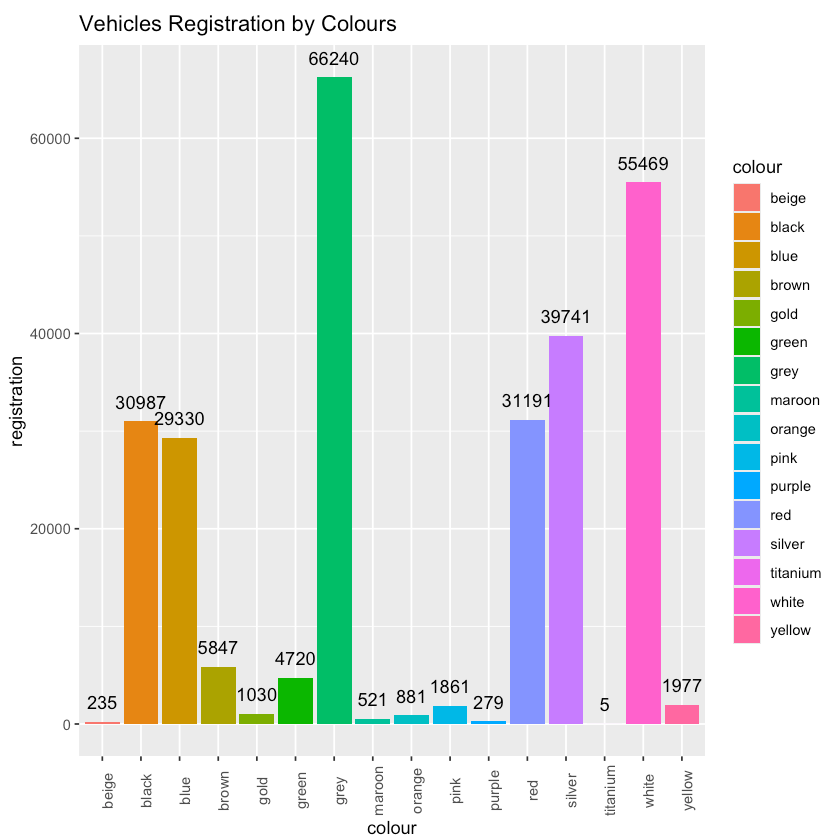
month	fuel	pct
<date>	<chr>	<dbl>
2026-01-01	diesel	0.26
2026-01-01	electric	9.18
2026-01-01	greendiesel	3.76
2026-01-01	hybrid_diesel	0.00
2026-01-01	hybrid_petrol	3.17
2026-01-01	petrol	83.63
2026-02-01	diesel	0.38
2026-02-01	electric	6.41
2026-02-01	greendiesel	5.04
2026-02-01	hybrid_petrol	3.90
2026-02-01	petrol	84.27
2026-03-01	diesel	0.28
2026-03-01	electric	6.95
2026-03-01	greendiesel	4.25
2026-03-01	hybrid_petrol	6.03
2026-03-01	petrol	82.47
2026-04-01	diesel	0.25
2026-04-01	electric	7.57
2026-04-01	greendiesel	3.88
2026-04-01	hybrid_diesel	0.01
2026-04-01	hybrid_petrol	6.35
2026-04-01	petrol	81.95



Petrol vehicles are still dominating the market but the electric and hybrid vehicles were steadily on the rise. People are now open to adapt newer, greener technology.

In [35]: *## By colour*

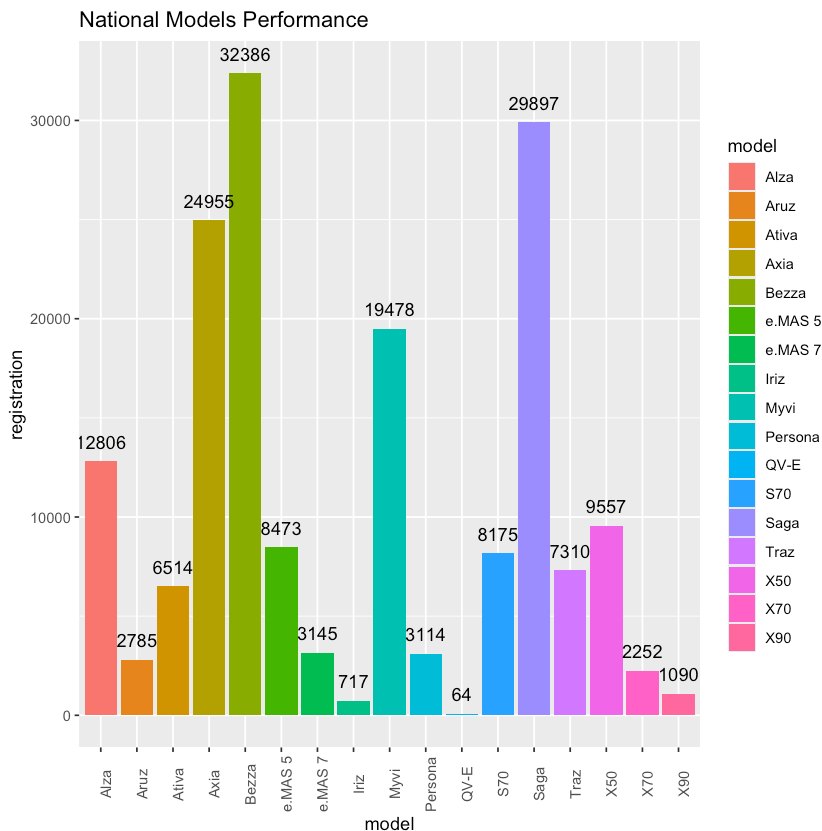
```
df_jpj26 %>%
  count(
    colour, name = "registration"
  ) %>%
  ggplot(aes(colour, registration, fill = colour)) +
  geom_col() +
  geom_text(aes(label = registration), vjust = -1) +
  theme(axis.text.x = element_text(angle = 90)) +
  ggtitle("Vehicles Registration by Colours")
```



Grey, white and silver tops the most preferred colour. Being a neutral colours its does look best in many situation and angle.

In [36]: *## Local brand performance*

```
df_jpj26 %>%
  filter(
    maker == 'Perodua' | maker == 'Proton'
  ) %>%
  count (
    model, name = "registration"
  ) %>%
  ggplot(aes(model, registration, fill = model)) +
  geom_col() +
  geom_text(aes(label = registration), vjust = -1 ) +
  theme(axis.text.x = element_text(angle = 90)) +
  ggtitle("National Models Performance")
```



Bezza and Saga tops the chart. Being the most affordable sedan its offer the best in terms of practicality, low fuel consumption and low maintenance its no brainer for most people.

Meanwhile Perodua QV-E scored lowest. Being Malaysia first homemade electric vehicle and only recently lauched. But many critics the price its offered not competetive enough when compare to other local EV such as e.Mas 5 and e.Mas 7 in terms of the interior cosmetics offering and not to mention, Perodua decision to create a subscription service for battery is highly controversial by many.

Let's move to study the electric and hybrid registration.

```
In [37]: ## EV registrations per month

df_ev26 <- df_jpj26 %>%
  filter(
    fuel == "electric"
  ) %>%
  mutate(
    month = month(date_reg, label = TRUE, abbr = TRUE)
  ) %>%
  count(
    month, name = "registrations"
  )

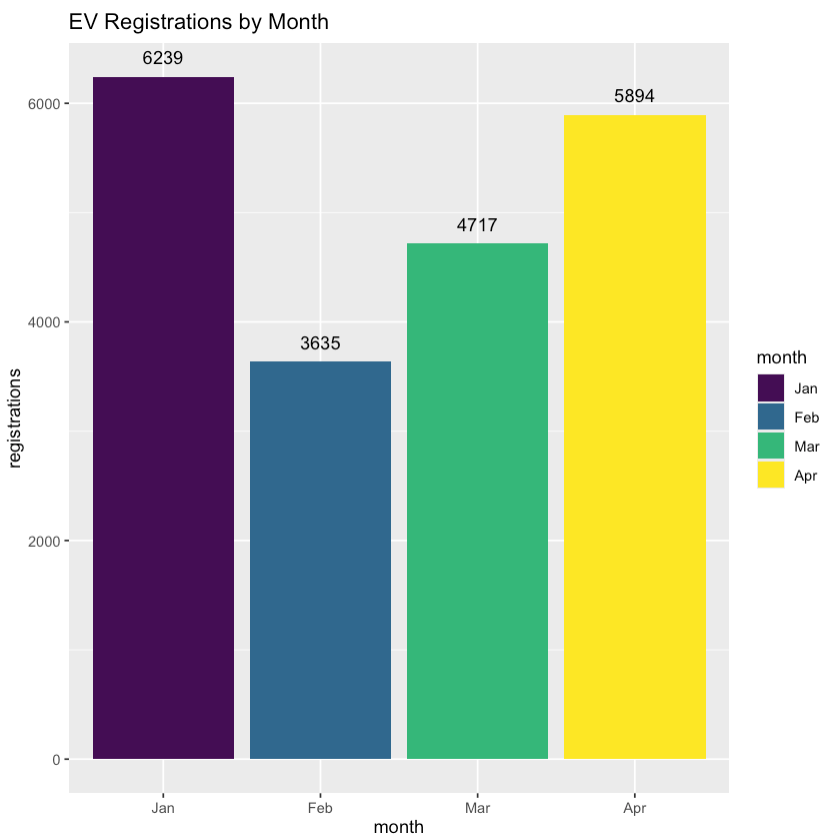
df_ev26
```

A tibble: 4 × 2

month **registrations**

<ord>	<int>
Jan	6239
Feb	3635
Mar	4717
Apr	5894

```
In [38]: df_ev26 %>% ggplot(aes(month, registrations, fill = month)) +
  geom_col() +
  geom_text(aes(label = registrations), vjust = -1) +
  ggtitle("EV Registrations by Month")
```



```
In [39]: ## Top 10 EV models registered.
```

```
df_jpj26 %>%
  filter(fuel == "electric") %>%
  group_by(maker, model) %>%
  summarise(
    total = n(), .groups = "drop"
  ) %>%
  arrange(desc(total)) %>%
  head(10)
```

A tibble: 10 × 3

maker	model	total
<chr>	<chr>	<int>
Proton	e.MAS 5	8473
BYD	Atto 3	1777
Proton	e.MAS 7	1390
Zeekr	7X	999
Chery	iCaur V23	972
BYD	Seal 6	669
BYD	Sealion	592
Chery	iCaur 03	577
Tesla	Model 3	459
MG	MG4	414

Proton e.Mas 5 is the cheapest EV car in available on the market today hence its top number 1 as it is affordable and suitable for people who just want to try the electric vehicle.

```
In [82]: ## EV registration by states excluding Rakan Niaga

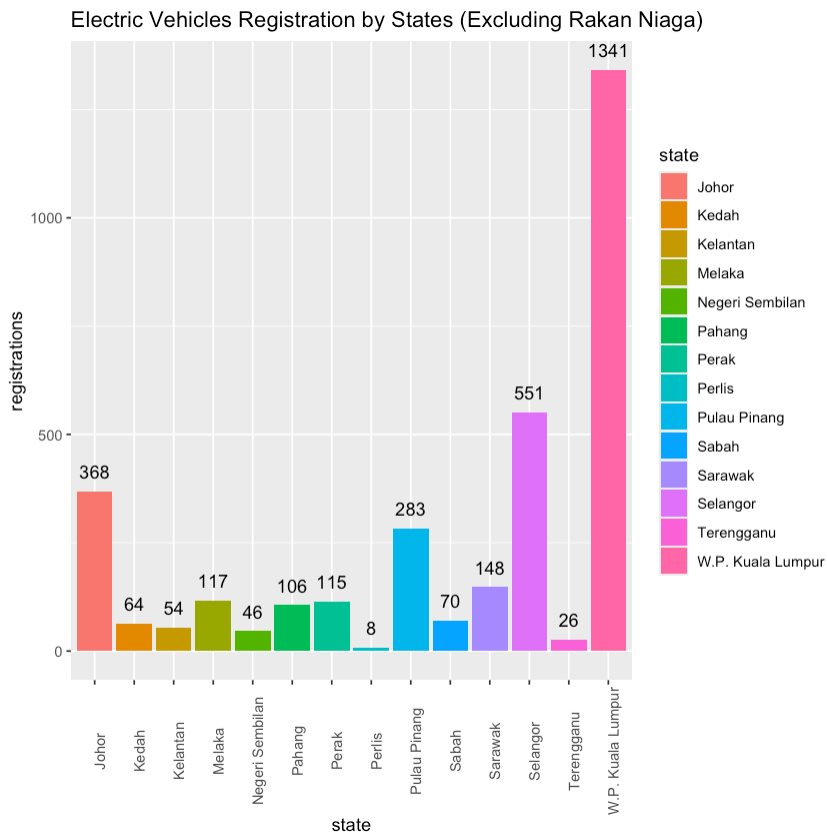
df_evstate26 <- df_jpj26 %>%
  filter(fuel == "electric", state != 'Rakan Niaga') %>%
  count(state, name = 'registrations', sort = TRUE)

df_evstate26

df_evstate26 %>%
  ggplot(aes(state, registrations, fill = state)) +
    geom_col() +
    geom_text(aes(label = registrations), vjust = -1) +
    theme(axis.text.x = element_text(angle = 90)) +
    ggtitle("Electric Vehicles Registration by States (Excluding Rakan N
```


A tibble: 14 × 2

state	registrations
<chr>	<int>
W.P. Kuala Lumpur	1341
Selangor	551
Johor	368
Pulau Pinang	283
Sarawak	148
Melaka	117
Perak	115
Pahang	106
Sabah	70
Kedah	64
Kelantan	54
Negeri Sembilan	46
Terengganu	26
Perlis	8



In [41]: ## Hybrid EV registrations per month

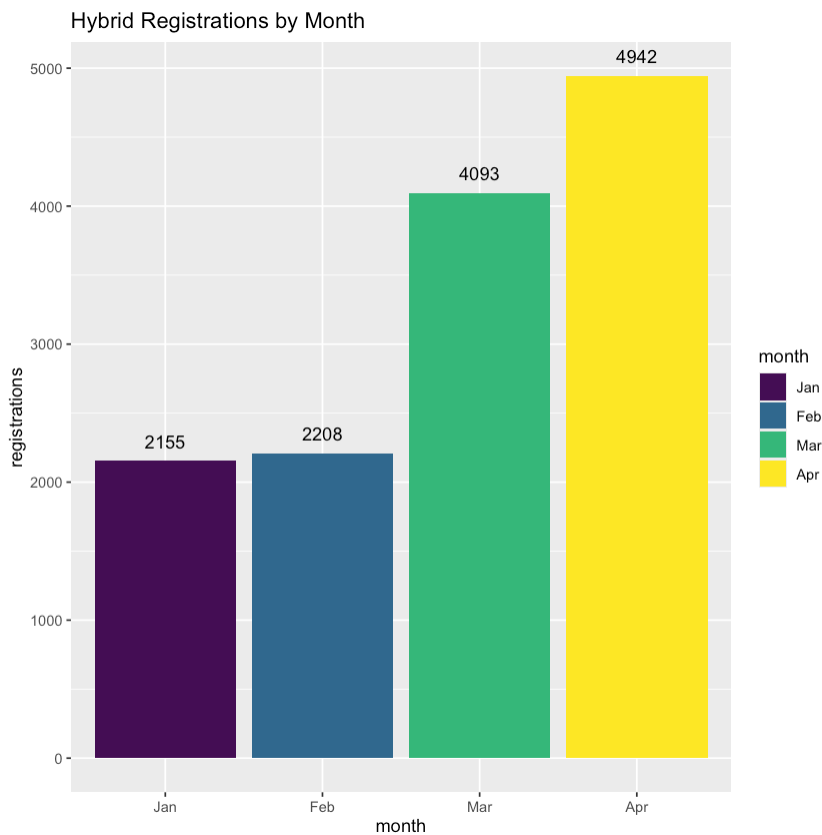
```
df_hev26 <- df_jpj26 %>%  
  filter(  
    fuel == "hybrid_petrol" | fuel == "hybrid_diesel"  
  ) %>%  
    mutate(  
      month = month(date_reg, label = TRUE, abbr = TRUE)  
    ) %>%  
      count(  
        month, name = "registrations"  
      )  
  
df_hev26
```

A tibble: 4 × 2

month registrations

<ord>	<int>
Jan	2155
Feb	2208
Mar	4093
Apr	4942

```
In [42]: df_hev26 %>% ggplot(aes(month, registrations, fill = month)) +  
  geom_col() +  
  geom_text(aes(label = registrations), vjust = -1) +  
  ggtitle("Hybrid Registrations by Month")
```



Hybrid electric vehicles has steady gain from January. Good performance.

In [43]: *## Top 10 hybrid electric models registered.*

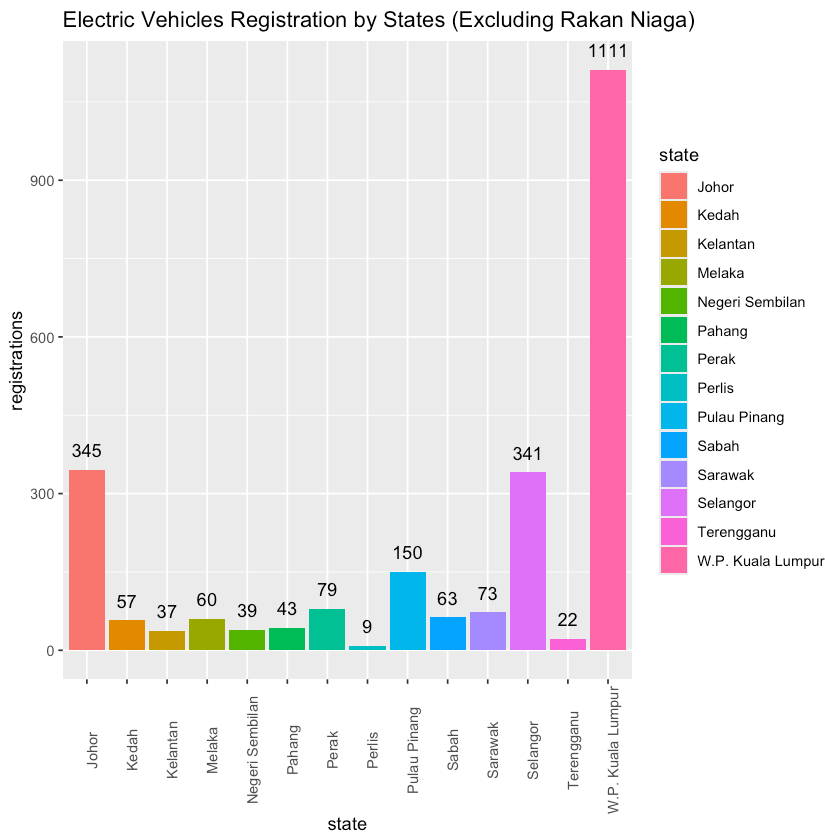
```
df_jpj26 %>%
  filter(fuel == "hybrid_petrol" | fuel == "hybrid_diesel") %>%
  group_by(maker, model) %>%
  summarise(
    total = n(), .groups = "drop"
  ) %>%
  arrange(desc(total)) %>%
  head(10)
```

A tibble: 10 × 3

maker	model	total
<chr>	<chr>	<int>
Toyota	Corolla Cross	1849
Proton	e.MAS 7	1755
Toyota	Vios	1386
Chery	Tiggo	1173
Great Wall	Haval H6	936
Honda	HR-V	827
Great Wall	Wey G9	610
Chery	Jaecoo J7	580
Honda	CR-V	563
Mercedes Benz	C-Class	446

In [44]: *## Hybrid registration by states excluding Rakan Niaga*

```
df_jpj26 %>%
  filter(fuel == "hybrid_petrol" | fuel == "hybrid_diesel", state != 'Rakan
count(state, name = 'registrations', sort = TRUE) %>%
ggplot(aes(state, registrations, fill = state)) +
  geom_col() +
  geom_text(aes(label = registrations), vjust = -1) +
  theme(axis.text.x = element_text(angle = 90)) +
  ggtitle("Electric Vehicles Registration by States (Excluding Rakan N
```



Overall, vehicles registrations in 2026 looks good. The rise of electric and hybrid vehicles registrations indicates people are now open to adapt the technology. If the makers continue to develop a more affordable vehicles, and government continue to support the growth of the industry, we may see its overtaken the petrol vehicles in no meantime. At the moment, whats hindered the wider adaptation are the lacks of infrastructure and the after sales support.

Let's compare the performance of the same period on 2025

```
In [45]: ## Importing 2025 dataset
```

```
df_jpj25 <- read.csv("cars_2025.csv")
```

```
In [46]: ## Inspecting the dataset
```

```
glimpse(df_jpj25)
```

```

Rows: 870,327
Columns: 7
$ date_reg <chr> "2025-01-01", "2025-01-01", "2025-01-01", "2025-01-01", "20
25...
$ type      <chr> "motokar", "window_van", "jip", "jip", "jip", "motokar", "m
ot...
$ maker     <chr> "BYD", "Cam", "Chery", "Chery", "Chery", "Dong Feng", "Dong
F...
$ model     <chr> "Seal", "Placer-X", "Jaecoo J7", "Jaecoo J7", "Tiggo", "Box
E...
$ colour    <chr> "white", "yellow", "green", "silver", "grey", "white", "pur
pl...
$ fuel      <chr> "electric", "greendiesel", "petrol", "petrol", "petrol", "e
le...
$ state     <chr> "Rakan Niaga", "Johor", "Rakan Niaga", "Rakan Niaga", "Raka
n ...

```

```
In [51]: ## Change date_reg to Date
```

```
df_jpj25$date_reg <- ymd(df_jpj25$date_reg)
```

```
In [52]: class(df_jpj25$date_reg)
```

```
'Date'
```

```
In [48]: ## checking for data integrity
```

```
## checking for null values
colSums(is.na(df_jpj25))
```

```
date_reg: 0 type: 0 maker: 0 model: 0 colour: 0 fuel: 0 state: 0
```

```
In [49]: ## Check duplication
```

```
sum(duplicated(df_jpj25))
```

```
708549
```

Again, we assuming there is no actual duplication in this data set. So we create registration ID to key to differentiate the registration process.

```
In [53]: ## Create Registration ID for year 2025
```

```
df_jpj25 <- df_jpj25 %>%
  group_by(date_reg) %>%
  mutate(
    regID = paste0(
      "reg_",
      format(date_reg, "%Y%m%d"),
      "_",
      sprintf("%04d", row_number())
    )
  ) %>%
  ungroup()
```

```
In [55]: head(df_jpj25)
         tail(df_jpj25)
```

A tibble: 6 × 8

date_reg	type	maker	model	colour	fuel	state	regID
<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
2025-01-01	motokar	BYD	Seal	white	electric	Rakan Niaga	reg_20250101_0001
2025-01-01	window_van	Cam	Placer-X	yellow	greendiesel	Johor	reg_20250101_0002
2025-01-01	jip	Chery	Jaecoo J7	green	petrol	Rakan Niaga	reg_20250101_0003
2025-01-01	jip	Chery	Jaecoo J7	silver	petrol	Rakan Niaga	reg_20250101_0004
2025-01-01	jip	Chery	Tiggo	grey	petrol	Rakan Niaga	reg_20250101_0005
2025-01-01	motokar	Dong Feng	Box E3	white	electric	Rakan Niaga	reg_20250101_0006

A tibble: 6 × 8

date_reg	type	maker	model	colour	fuel	state	regID
<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
2025-12-31	jip	Zeekr	7X	grey	electric	Rakan Niaga	reg_20251231_15365
2025-12-31	jip	Zeekr	7X	white	electric	Selangor	reg_20251231_15366
2025-12-31	jip	Zeekr	7X	grey	electric	W.P. Kuala Lumpur	reg_20251231_15367
2025-12-31	jip	Zeekr	7X	green	electric	Selangor	reg_20251231_15368
2025-12-31	jip	Zeekr	7X	white	electric	Rakan Niaga	reg_20251231_15369
2025-12-31	jip	Zeekr	X	white	electric	Rakan Niaga	reg_20251231_15370

This 2025 data set was a full year registration. We need to remove the registration after the April 2025 in order to compare the same period with the previous 2026.

```
In [56]: ## Remove rows after 30 April 2025

df_jpj25 <- subset(df_jpj25, date_reg < as.Date("2025-05-01"))
```

```
In [57]: tail(df_jpj25)
```

A tibble: 6 × 8

date_reg	type	maker	model	colour	fuel	state	regID
<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
2025-04-30	window_van	Zeekr	009	white	electric	Rakan Niaga	reg_20250430_14559
2025-04-30	window_van	Zeekr	009	black	electric	W.P. Kuala Lumpur	reg_20250430_14560
2025-04-30	window_van	Zeekr	009	white	electric	Rakan Niaga	reg_20250430_14561
2025-04-30	window_van	Zeekr	009	black	electric	W.P. Kuala Lumpur	reg_20250430_14562
2025-04-30	window_van	Zeekr	009	white	electric	W.P. Kuala Lumpur	reg_20250430_14563
2025-04-30	window_van	Zeekr	009	black	electric	Rakan Niaga	reg_20250430_14564

In [58]: *## Count car registration for same period in 2025*

```
count(df_jpj25, name = "total registration")
```

A tibble: 1 × 1

total registration

<int>

263578

In [103... *## Comparing 2025 and 2026*

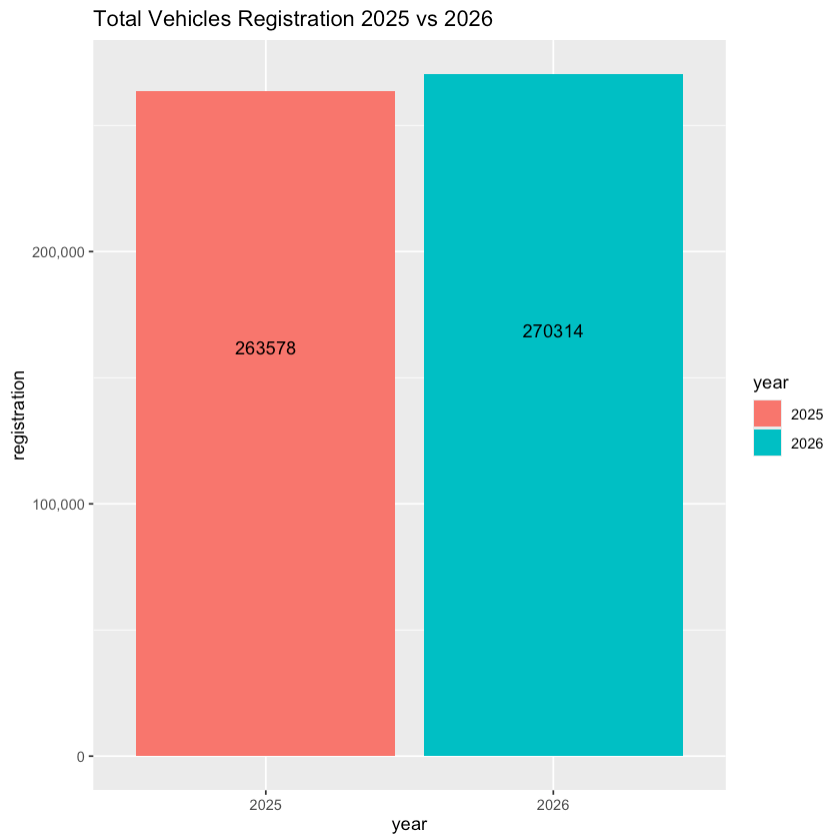
```
df_vs <- bind_rows(
  summarise(df_jpj25, year = "2025", registration = n()),
  summarise(df_jpj26, year = "2026", registration = n())
)

df_vs %>%
  mutate(
    pct_diff = round((registration - lag(registration)) / lag(registration))
  )

df_vs %>% ggplot(aes(x = year, y = registration, fill = year)) +
  geom_col() +
  scale_y_continuous(labels = comma) +
  geom_text(aes(label = registration), vjust = 20) +
  ggtitle("Total Vehicles Registration 2025 vs 2026")
```

A tibble: 2 × 3

year	registration	pct_diff
<chr>	<int>	<dbl>
2025	263578	NA
2026	270314	2.56



During the same period, 2026 recording a 2.56% increase in registration compare to 2025.

In [61]: *## 2025 v 2026 Month to month comparison*

```
summary25 <- df_jpj25 %>%
  mutate(month = month(date_reg, label = TRUE, abbr = TRUE)) %>%
  group_by(month) %>%
  summarise("2025" = n(), .groups = "drop")

summary26 <- df_jpj26 %>%
  mutate(month = month(date_reg, label = TRUE, abbr = TRUE)) %>%
  group_by(month) %>%
  summarise("2026" = n(), .groups = "drop")
```

In [63]:

```
compare_month <- summary25 %>%
  left_join(summary26, by = "month") %>%
  mutate(
    pct = round(((`2026` - `2025`) / `2025`) * 100, 2)
  )
```

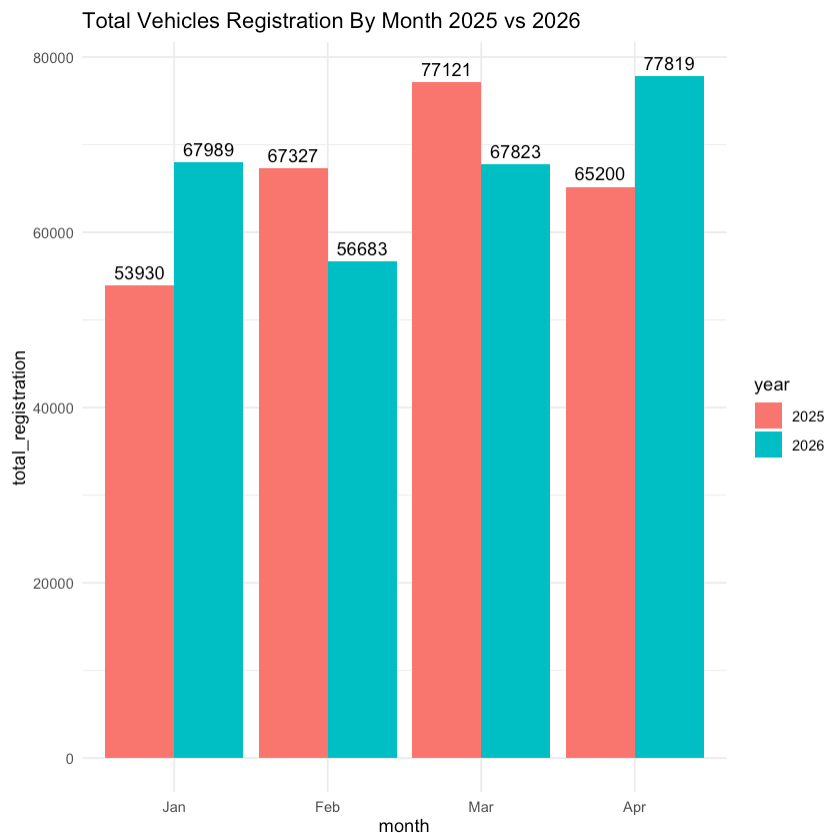


```
In [98]: compare_month

compare_month %>%
  pivot_longer(
    cols = c("2025", "2026"),
    names_to = "year",
    values_to = "total_registration"
  ) %>%
  ggplot(aes(x = month, y = total_registration, fill = year)) +
  geom_col(position = "dodge") +
  geom_text(aes(label = total_registration), position = position_dodge(wid
  theme_minimal() +
  ggtitle("Total Vehicles Registration By Month 2025 vs 2026")
```

A tibble: 4 × 4

month	2025	2026	pct
<ord>	<int>	<int>	<dbl>
Jan	53930	67989	26.07
Feb	67327	56683	-15.81
Mar	77121	67823	-12.06
Apr	65200	77819	19.35



Although number of registration in 2026 is higher overall, month to month performance shows a different story. In 2025, vehicles registrations rise steadily from January to March then drop significantly in April. For 2026, registration slow down on February but

rising again onwards.

Let's study the fuel types vehicles trend.

```
In [ ]: fuel_month25 <- df_jpj25 %>%
  mutate(month = floor_date(date_reg, "month")) %>%
  count(month, fuel) %>%
  group_by(month) %>%
  mutate(pct = round(n / sum(n) * 100, 2)) %>%
  ungroup() %>%
  select(-n)
```

fuel_month25

```
In [ ]: ## Combined fuel_month & fuel_month25
```

```
fuel_monthvs <- bind_rows(
  fuel_month25, fuel_month
)
```

fuel_monthvs

```
In [ ]: ## Create new column to simplify plotting with ggplot
```

```
fuel_monthvsplot <- fuel_monthvs %>%
  mutate(
    year = year(month), month_num = month(month),
    month_name = factor( month(month, label = TRUE, abbr = TRUE),
      levels = month.abb[1:4] ),
    x = month_num + if_else(year == 2025, -0.2, 0.2) )
```

fuel_monthvsplot

```
In [69]: ## Finding difference between 2025 and 2026
```

```
fuel_monthvsDiff <- fuel_monthvsplot %>%
  select(month_name, fuel, year, pct) %>%
  pivot_wider(
    names_from = year,
    values_from = pct
  ) %>%
  mutate(
    difference = `2026` - `2025`
  )
```

```
## Remove hybrid diesel because it is not significant
```

```
fuel_monthvsDiff <- fuel_monthvsDiff %>%
  filter(fuel != "hybrid_diesel")
```

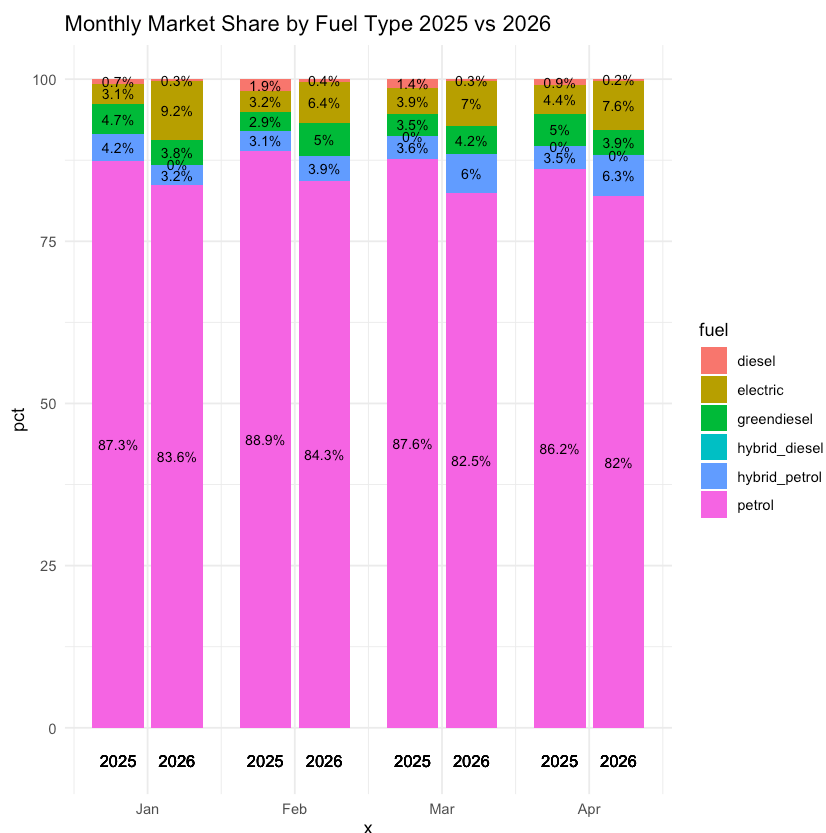
fuel_monthvsDiff

```
## Plotting
```

```
ggplot(
  fuel_monthvsplot, aes( x = x, y = pct, fill = fuel ) ) +
  geom_col( width = 0.35 ) +
  geom_text(aes(label = paste0(round(pct, 1), "%")), position = position_s
  geom_text(aes(x = x, y = -5, label = year), inherit.aes = FALSE, size =
  scale_x_continuous( breaks = 1:4, labels = month.abb[1:4] ) +
  ggtitle("Monthly Market Share by Fuel Type 2025 vs 2026") +
  theme_minimal()
```

A tibble: 20 × 5

month_name	fuel	2025	2026	difference
<ord>	<chr>	<dbl>	<dbl>	<dbl>
Jan	diesel	0.69	0.26	-0.43
Jan	electric	3.14	9.18	6.04
Jan	greendiesel	4.68	3.76	-0.92
Jan	hybrid_petrol	4.17	3.17	-1.00
Jan	petrol	87.32	83.63	-3.69
Feb	diesel	1.88	0.38	-1.50
Feb	electric	3.21	6.41	3.20
Feb	greendiesel	2.85	5.04	2.19
Feb	hybrid_petrol	3.14	3.90	0.76
Feb	petrol	88.92	84.27	-4.65
Mar	diesel	1.44	0.28	-1.16
Mar	electric	3.86	6.95	3.09
Mar	greendiesel	3.48	4.25	0.77
Mar	hybrid_petrol	3.59	6.03	2.44
Mar	petrol	87.63	82.47	-5.16
Apr	diesel	0.86	0.25	-0.61
Apr	electric	4.44	7.57	3.13
Apr	greendiesel	5.02	3.88	-1.14
Apr	hybrid_petrol	3.48	6.35	2.87
Apr	petrol	86.20	81.95	-4.25



Comparing 2025 and 2026, petrol and diesel powered vehicles sees a decline in registrations while the rest saw a significant jump especially the electric vehicles.

Now, we comparing the registrations by states 2025 vs 2026.

```
In [70]: df_states26 <- df_jpj26 %>%
  filter(state != 'Rakan Niaga') %>%
  count (state, name = "registrations")

df_states25 <- df_jpj25 %>%
  filter(state != 'Rakan Niaga') %>%
  count (state, name = "registrations")

df_statesvs <- bind_rows(
  df_states25 %>% mutate(year = "2025"),
  df_states26 %>% mutate(year = "2026")
)

df_statesvs <- df_statesvs %>%
  select(state, year, registrations) %>%
  pivot_wider(
    names_from = year,
    values_from = registrations
  ) %>%
  mutate(
    pct = round((`2026` - `2025`) / `2025` * 100, 2)
  ) %>%
  arrange(desc(pct))
```

```

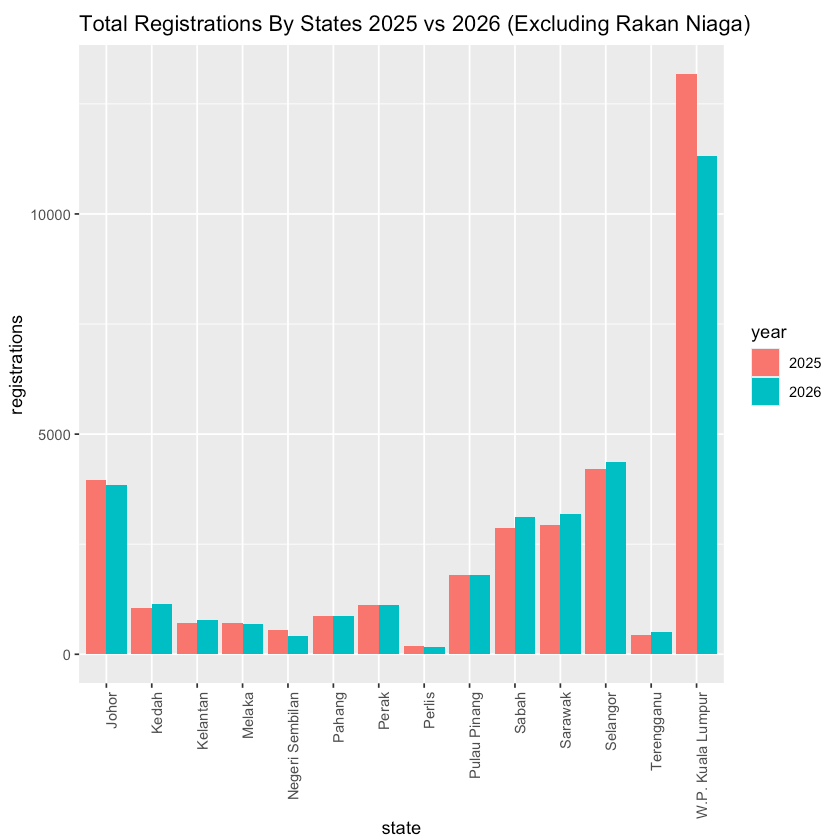
df_statesvs

df_statesvs %>%
  pivot_longer(
    cols = c("2025", "2026"),
    names_to = "year",
    values_to = "registrations"
  ) %>%
  ggplot(aes(x = state, y = registrations, fill = year)) +
    geom_col(position = "dodge") +
    theme(
      axis.text.x = element_text(angle = 90, hjust = 1)
    ) +
    ggtitle("Total Registrations By States 2025 vs 2026 (Excluding Rakan N

```

A tibble: 14 × 4

	state	2025	2026	pct
	<chr>	<int>	<int>	<dbl>
	Terengganu	432	498	15.28
	Kelantan	714	785	9.94
	Sarawak	2938	3195	8.75
	Sabah	2877	3117	8.34
	Kedah	1057	1132	7.10
	Selangor	4205	4376	4.07
	Pahang	860	873	1.51
	Pulau Pinang	1801	1811	0.56
	Perak	1117	1111	-0.54
	Melaka	709	694	-2.12
	Johor	3966	3838	-3.23
	Perlis	189	163	-13.76
	W.P. Kuala Lumpur	13178	11323	-14.08
	Negeri Sembilan	554	414	-25.27



Negeri Sembilan recorded a highest decline in registrations while registration in Kuala Lumpur in 2026 also decline by a margin eventhough its still top the list.

In [71]: ## EV for 2025

```
df_ev25 <- df_jpj25 %>%
  filter(
    fuel == "electric"
  ) %>%
  mutate(
    month = month(date_reg, label = TRUE, abbr = TRUE)
  ) %>%
  count(
    month, name = "registrations"
  )
df_ev25
```

A tibble: 4 × 2

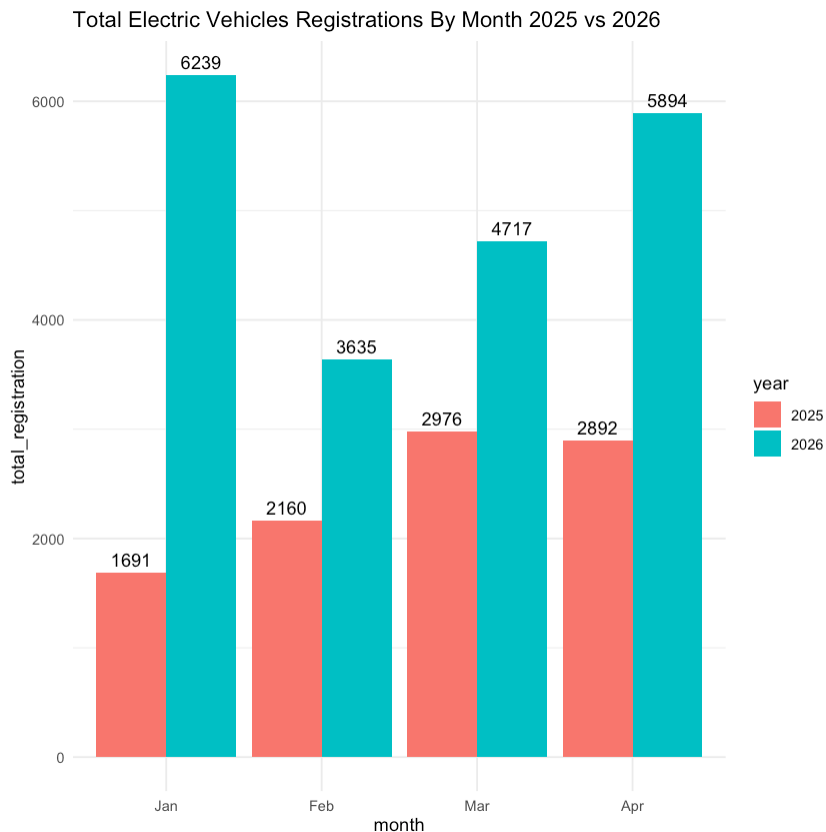
month	registrations
<ord>	<int>
Jan	1691
Feb	2160
Mar	2976
Apr	2892

In [97]: `## Plotting EV 2025 with EV 2026`

```
df_ev_vs <- df_ev25 %>%
  left_join(df_ev26, by = "month")

df_ev_vs <- df_ev_vs %>%
  rename(
    `2025` = registrations.x,
    `2026` = registrations.y
  )

df_ev_vs %>%
  pivot_longer(
    cols = c("2025", "2026"),
    names_to = "year",
    values_to = "total_registration"
  ) %>%
  ggplot(aes(x = month, y = total_registration, fill = year)) +
  geom_col(position = "dodge") +
  geom_text(aes(label = total_registration), position = position_dodge(wid
  theme_minimal() +
  ggtitle("Total Electric Vehicles Registrations By Month 2025 vs 2026")
```



In [95]: `## States EV registration 2025 vs 2026`

```
df_evstate25 <- df_jpj25 %>%
  filter(fuel == "electric", state != 'Rakan Niaga') %>%
  count(state, name = 'registrations', sort = TRUE)
```

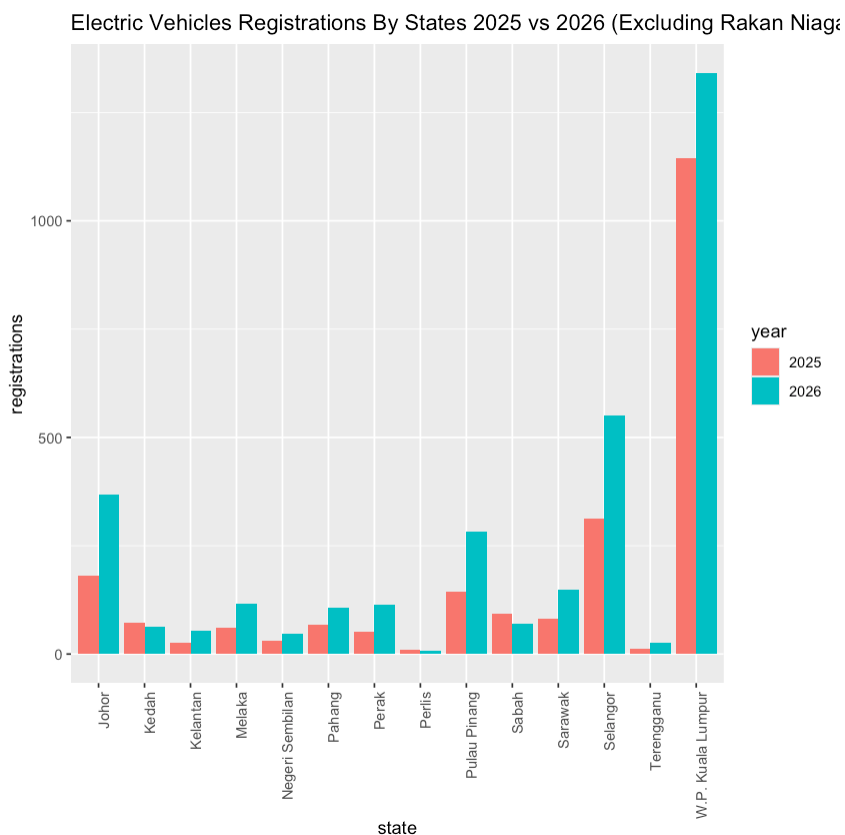
```
## Join dataframe

df_evstate_vs <- df_evstate25 %>%
  left_join(df_evstate26, by = "state")

df_evstate_vs <- df_evstate_vs %>%
  rename(
    `2025` = registrations.x,
    `2026` = registrations.y
  )

## Plotting

df_evstate_vs %>%
  pivot_longer(
    cols = c("2025", "2026"),
    names_to = "year",
    values_to = "registrations"
  ) %>%
  ggplot(aes(x = state, y = registrations, fill = year)) +
  geom_col(position = "dodge") +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1)
  ) +
  ggtitle("Electric Vehicles Registrations By States 2025 vs 2026 (Exclu
```



Demand for electric vehicles in 2026 was very strong with a very significant jump of registrations recorded. Every states recording a jump in registrations except for Sabah.

Government ev vehicles tax incentive, more affordable models in the market, and people finally adapting to the tech definitely contribute to the numbers.

In conclusion, the number of vehicles registration remains positive in 2026, even stronger against 2025 on the same period despite current economics challenge. In 2026 we can see the rise of electric and hybrid vehicles against the traditional petrol and diesel, which tell us people now are open to move away from the traditional fossil fuel powered vehicles towards a greener, energy efficient vehicles. A challenging ongoing fuel crisis and the government initiative to push the technology, people today are more confident to adapt. Vehicles market today are flooded with many options to choose to suits many level of income group.

What hindered a widespread nationwide adoption of electric and hybrid vehicles is the lack of charging infrastructure. Stakeholder and regulators should engage more to improve the charging facilities. If this main issue is tackled, more and more electric and hybrid vehicles will be registered from states like Perlis, Terengganu or Kelantan.